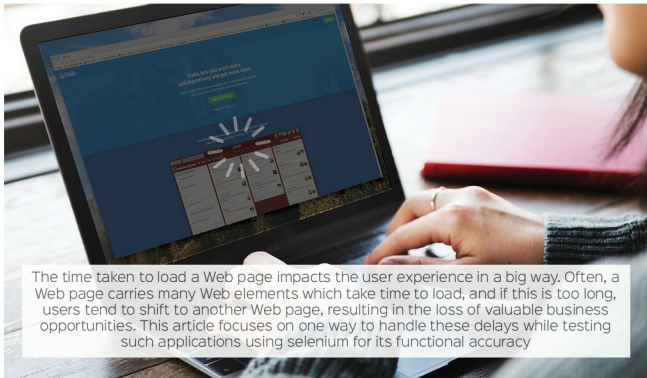


The Long and Short of Waits in Web Applications Using Selenium



The time taken to load a Web page impacts the user experience in a big way. Often, a Web page carries many Web elements which take time to load, and if this is too long, users tend to shift to another Web page, resulting in the loss of valuable business opportunities. This article focuses on one way to handle these delays while testing such applications using selenium for its functional accuracy

Most Web applications we deal with are developed using Ajax and JavaScript. In such cases, when a page is loaded by the browser, the Web elements will take some time to load and the *loading time* can be different in each run. This leads to different exceptions, resulting in a report with the wrong results even though the Web app is working correctly.

Handling the delays with accuracy

On noticing a delay in the appearance of a UI element on the page, let's suppose we use `Thread.sleep()` to handle it. The next time we run the test, the UI element loads instantly. But what about `Thread.sleep()`, which we introduced into our code? It still halts the execution, thereby inducing the extra wait time even though it's not required. Finally, executing our entire test suite takes its own sweet time to finish. Though the results are accurate and reliable, in a practical world, this approach makes no sense when manual intervention testing of the application gets the job done in one minute, compared to our automated script, which takes three minutes due to the extra overhead delay of `sleep()` that we introduced.

So to make automation robust and accurate, as well as to synchronise our test tool with the application under test by handling these delays dynamically, we use *waits* provided by Selenium.

Types of waits

- **Implicit wait:** An implicit wait is to instruct WebDriver to poll the DOM for a certain amount of time when trying to find an element or elements, if these are not immediately available. The default setting is 0. Once set, the implicit wait is set for the life of the WebDriver object instance.

Declaration:

```
driver.manage().timeouts().implicitlyWait(10, TimeUnit.SECONDS);
```

When to use: Not recommended, but better than using `Thread.sleep()`.

- **PageLoadTimeout wait:** This wait is focused on the time a Web page needs to be loaded – the `pageLoadTimeout` limits the time that the script allots for a Web page to be displayed before proceeding with other operations on the Web elements present on the page. If the page loads within the time, then the script continues. If the page does not load within the timeout, the script will be stopped by a `TimeoutException`.